

500 MCQ OF OS

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✓ 1–50 MCQ Questions on Basics of OS

1. **Which of the following is not a function of the Operating System?**

- a) Memory management
- b) File management
- c) Compiler design
- d) Process management

✓ **Answer: c**

2. **The core of the operating system is called:**

- a) Shell
- b) Kernel
- c) Command
- d) Process

✓ **Answer: b**

3. **Which of the following is a type of OS?**

- a) Batch OS
- b) Compiler OS
- c) Interpreter OS
- d) None

✓ **Answer: a**

4. **Time-sharing systems are also known as:**

- a) Multitasking systems
- b) Real-time systems
- c) Distributed systems
- d) Batch systems

✓ **Answer: a**

5. **Which of these is not a real-time OS?**

- a) RTLinux
- b) Windows 10
- c) VxWorks
- d) QNX

✓ **Answer: b**

6. **An Operating System acts as:**

- a) User
- b) Interface between user and hardware
- c) Hardware
- d) Software only

✓ **Answer: b**

7. **Which is not a component of the OS?**
a) Process Management
b) Word Processor
c) File System
d) I/O Management
✓ **Answer: b**
8. **Which scheduling algorithm is non-preemptive?**
a) Round Robin
b) Shortest Remaining Time
c) FCFS
d) Priority (with preemption)
✓ **Answer: c**
9. **The purpose of an interrupt is to:**
a) Slow down CPU
b) Let the CPU know that a device needs attention
c) Terminate program
d) Allocate memory
✓ **Answer: b**
10. **What is multitasking?**
a) Running multiple tasks by multiple users
b) Running one task for multiple users
c) Running multiple tasks simultaneously by one user
d) Running one task by one user
✓ **Answer: c**
11. **Which is a multi-user OS?**
a) MS-DOS
b) Windows 98
c) UNIX
d) Windows XP
✓ **Answer: c**
12. **Which of the following is a distributed OS?**
a) MS-DOS
b) Windows NT
c) Amoeba
d) Windows XP
✓ **Answer: c**
13. **Shell is:**
a) Hardware
b) Interface between user and kernel
c) Part of CPU
d) Input device
✓ **Answer: b**
14. **Which of the following is not a valid process state?**
a) New
b) Running
c) Waiting
d) Termination-ready
✓ **Answer: d**

15. **The main function of the command interpreter is:**
- a) Manage processes
 - b) Translate commands into machine language
 - c) Get and execute the next command
 - d) Handle I/O
- ✓ **Answer: c**
16. **Bootting is a process of:**
- a) Deleting files
 - b) Restarting the system
 - c) Loading the OS into main memory
 - d) Formatting the hard drive
- ✓ **Answer: c**
17. **Which of these is a network OS?**
- a) UNIX
 - b) Windows Server
 - c) MS-DOS
 - d) None
- ✓ **Answer: b**
18. **Which of the following is not a characteristic of a real-time OS?**
- a) Fast response
 - b) Predictable performance
 - c) Large memory requirement
 - d) Deterministic behavior
- ✓ **Answer: c**
19. **UNIX OS is written in which language?**
- a) C
 - b) C++
 - c) Java
 - d) Assembly
- ✓ **Answer: a**
20. **The part of the OS that remains in memory while the system is running is:**
- a) Shell
 - b) Kernel
 - c) Compiler
 - d) Loader
- ✓ **Answer: b**
21. **Which of the following is not an OS?**
- a) Windows
 - b) Linux
 - c) Oracle
 - d) Android
- ✓ **Answer: c**
22. **Main goal of an OS is:**
- a) Security
 - b) Convenience
 - c) Efficient resource management
 - d) All of the above
- ✓ **Answer: d**

23. **The first program run on a computer when it boots up is:**

- a) OS
- b) Shell
- c) BIOS
- d) Bootloader

✓ **Answer: d**

24. **Which is the most important part of a mobile OS?**

- a) GUI
- b) Kernel
- c) File Manager
- d) Application

✓ **Answer: b**

25. **Which OS is used in Apple's iPhone?**

- a) Android
- b) Symbian
- c) iOS
- d) UNIX

✓ **Answer: c**

26. **Which of these is not an open-source OS?**

- a) Linux
- b) Windows
- c) FreeBSD
- d) Ubuntu

✓ **Answer: b**

27. **Which is not a task of OS in relation to memory?**

- a) Allocation
- b) Deallocation
- c) Compilation
- d) Protection

✓ **Answer: c**

28. **Which file contains the settings and configuration of OS?**

- a) Binary file
- b) Config file
- c) System file
- d) Log file

✓ **Answer: b**

29. **BIOS stands for:**

- a) Binary Input Output System
- b) Basic Integrated Operating System
- c) Basic Input Output System
- d) None of these

✓ **Answer: c**

30. **The OS manages hardware via:**

- a) Device drivers
- b) Application
- c) Compilers
- d) Editors

✓ **Answer: a**

31. **Which of these is not part of OS architecture?**
a) Layered
b) Monolithic
c) Microkernel
d) Macroeditor
✓ **Answer: d**
32. **Which OS was developed first in the history of computers?**
a) UNIX
b) MS-DOS
c) GMOS
d) MacOS
✓ **Answer: c**
33. **System calls are used for:**
a) Communication between user and kernel
b) File transfer
c) Network communication
d) Game execution
✓ **Answer: a**
34. **Which of the following is an interface between hardware and software?**
a) User
b) OS
c) Compiler
d) Assembler
✓ **Answer: b**
35. **Which feature allows OS to run multiple programs at once?**
a) Compilation
b) Translation
c) Multitasking
d) Fragmentation
✓ **Answer: c**
36. **What kind of software is the OS?**
a) System software
b) Application software
c) Utility software
d) Development software
✓ **Answer: a**
37. **Which OS has GUI?**
a) MS-DOS
b) Linux
c) Windows
d) Both b and c
✓ **Answer: d**
38. **Which type of OS is used in aircraft and life-saving devices?**
a) Real-time
b) Distributed
c) Network
d) Batch
✓ **Answer: a**

39. In which OS type does each user have its own terminal and memory space?
- a) Multi-user
 - b) Time-sharing
 - c) Single-user
 - d) Real-time
- ✓ Answer: a
40. File system is responsible for:
- a) Naming and storing files
 - b) Formatting disk
 - c) Installing software
 - d) RAM management
- ✓ Answer: a
41. Which one is an example of a batch system?
- a) Unix
 - b) MS-DOS
 - c) IBM OS/360
 - d) Android
- ✓ Answer: c
42. Which OS provides tight deadline-based responses?
- a) Real-time OS
 - b) Multi-user OS
 - c) Batch OS
 - d) Network OS
- ✓ Answer: a
43. Spooling is useful for:
- a) Managing CPU
 - b) Managing disk
 - c) Managing I/O
 - d) Managing cache
- ✓ Answer: c
44. Which OS component schedules tasks?
- a) Scheduler
 - b) Compiler
 - c) Loader
 - d) Linker
- ✓ Answer: a
45. Which is not a type of system call?
- a) Process control
 - b) File manipulation
 - c) Device management
 - d) Gaming
- ✓ Answer: d
46. Which interrupt is not maskable?
- a) Hardware interrupt
 - b) Software interrupt
 - c) Non-maskable interrupt (NMI)
 - d) All of the above
- ✓ Answer: c

47. **OS security is mainly implemented via:**

- a) Cryptography
- b) Hardware lock
- c) System calls
- d) Access control

✓ **Answer: d**

48. **Which of the following is not a mobile OS?**

- a) iOS
- b) Android
- c) Linux Mint
- d) HarmonyOS

✓ **Answer: c**

49. **Which of the following manages execution of multiple jobs?**

- a) Scheduler
- b) Loader
- c) Compiler
- d) Editor

✓ **Answer: a**

50. **Which component loads the OS during boot?**

- a) BIOS
- b) Kernel
- c) Bootloader
- d) File system

✓ **Answer: c**

✓ **51–100 MCQs on Process & Thread Management**

51. **Which of the following is not a process state?**

- a) New
- b) Ready
- c) Waiting
- d) Termination-ready

✓ **Answer: d**

52. **A thread shares which of the following with peer threads in the same process?**

- a) Registers
- b) Stack
- c) Code segment
- d) Program counter

✓ **Answer: c**

53. **Which system call is used to create a new process in Unix?**

- a) exec()
- b) fork()
- c) create()
- d) new()

✓ **Answer: b**

54. **The parent process uses which system call to replace its address space with a new program?**

- a) replace()
- b) exec()

- c) fork()
- d) exit()

✓ **Answer: b**

55. **Which process is created when a new process is created by fork()?**

- a) Child
- b) Parent
- c) Sibling
- d) Thread

✓ **Answer: a**

56. **Context switching means:**

- a) Changing the CPU from one process to another
- b) Switching between memory segments
- c) Swapping disk blocks
- d) Flushing cache

✓ **Answer: a**

57. **What is the degree of multiprogramming?**

- a) Number of processes in memory
- b) Number of CPU cycles
- c) Number of interrupts
- d) Number of users

✓ **Answer: a**

58. **The Process Control Block (PCB) does not contain:**

- a) Process ID
- b) Program counter
- c) Page table
- d) User credentials

✓ **Answer: d**

59. **Which is not a thread type?**

- a) User-level thread
- b) Kernel-level thread
- c) Hybrid thread
- d) Stackless thread

✓ **Answer: d**

60. **Which scheduler selects from among the processes in memory, ready to execute?**

- a) Long-term scheduler
- b) Mid-term scheduler
- c) Short-term scheduler
- d) I/O scheduler

✓ **Answer: c**

61. **Which of the following does not interrupt a running process?**

- a) Timer
- b) Scheduler process
- c) I/O completion
- d) System call

✓ **Answer: d**

62. **Which one is not a benefit of threads over processes?**

- a) Faster context switching
- b) Easier inter-thread communication
- c) More memory usage

- d) Responsiveness
✓ **Answer: c**
63. **Multithreading improves:**
a) Disk speed
b) CPU utilization
c) I/O speed
d) Compilation
✓ **Answer: b**
64. **When a process is swapped out, its PCB is:**
a) Deleted
b) Saved in memory
c) Saved on disk
d) Ignored
✓ **Answer: c**
65. **Which of the following creates zombie processes?**
a) Orphan process
b) Parent terminating early
c) Child terminating but parent not collecting exit status
d) Using exec() after fork()
✓ **Answer: c**
66. **When two threads access shared data, what should be used?**
a) Page table
b) Mutex
c) Loader
d) PCB
✓ **Answer: b**
67. **Which state is entered after the process finishes execution?**
a) Waiting
b) Running
c) Terminated
d) Ready
✓ **Answer: c**
68. **Which is true about a thread?**
a) It has its own stack
b) It has its own PCB
c) It doesn't share memory
d) It requires its own kernel
✓ **Answer: a**
69. **A process that has terminated but whose parent has not waited for it is known as:**
a) Orphan
b) Zombie
c) Daemon
d) Idle
✓ **Answer: b**
70. **Which of the following is NOT shared between threads?**
a) Code
b) Stack
c) Data

d) Files

✓ **Answer: b**

71. **Threads can't run concurrently on a single-core system without:**

- a) Compiling
- b) Interrupts
- c) Scheduling
- d) Fork

✓ **Answer: c**

72. **What is the result of calling fork() once in a program?**

- a) Only parent process continues
- b) Only child continues
- c) Two processes continue
- d) Program ends

✓ **Answer: c**

73. **Which one of the following is responsible for context switching?**

- a) Interrupts
- b) Compiler
- c) Linker
- d) Loader

✓ **Answer: a**

74. **A process may be blocked if:**

- a) It finishes execution
- b) It is executing
- c) It is waiting for an I/O
- d) It is loaded

✓ **Answer: c**

75. **What is the process of starting a new thread called?**

- a) Execution
- b) Forking
- c) Spawning
- d) Splitting

✓ **Answer: c**

76. **Which part of a process is read-only?**

- a) Stack
- b) Heap
- c) Code
- d) Data

✓ **Answer: c**

77. **Which one is NOT a valid reason for process termination?**

- a) Normal exit
- b) Timeout
- c) Waiting for I/O
- d) Parent request

✓ **Answer: c**

78. **What is orphan process?**

- a) A process with multiple parents
- b) A child process whose parent has terminated
- c) A parent process that spawns no child

d) A process stuck in deadlock

✓ **Answer: b**

79. **Which thread model maps many user threads to many kernel threads?**

a) 1:1

b) M:1

c) M:N

d) N:1

✓ **Answer: c**

80. **A process is in waiting state when:**

a) It's using CPU

b) It's ready to execute

c) Waiting for I/O

d) Terminated

✓ **Answer: c**

81. **Threads are often used in:**

a) Word processing

b) Web servers

c) Compilers

d) Assemblers

✓ **Answer: b**

82. **The execution context of a thread does not include:**

a) Stack pointer

b) Program counter

c) Scheduling queue

d) Process ID

✓ **Answer: d**

83. **What does `exec()` do?**

a) Create new thread

b) Terminate thread

c) Replace current process memory with a new program

d) Copy process

✓ **Answer: c**

84. **Which scheduling algorithm is best for time-sharing systems?**

a) FCFS

b) SJF

c) Round Robin

d) Priority

✓ **Answer: c**

85. **Which of these uses the most context switches?**

a) Batch system

b) Real-time OS

c) Multitasking system

d) Single-user OS

✓ **Answer: c**

86. **Threads provide better performance because:**

a) They use less memory

b) They don't need a compiler

c) No need for code

d) They run in parallel always

✓ **Answer: a**

87. **How many threads minimum does a Java program have?**

a) 0

b) 1

c) 2

d) Unlimited

✓ **Answer: c** (main thread + garbage collector)

88. **What is the function of the loader?**

a) Allocates memory

b) Loads executable into memory

c) Schedules processes

d) Terminates processes

✓ **Answer: b**

89. **If no scheduling is done, what happens?**

a) All processes run

b) No process runs

c) Only one process runs until it finishes

d) Processes run randomly

✓ **Answer: c**

90. **Process synchronization is used to:**

a) Avoid race conditions

b) Manage memory

c) Load processes

d) Link objects

✓ **Answer: a**

91. **The role of dispatcher is to:**

a) Handle interrupts

b) Assign CPU to selected process

c) Remove completed processes

d) Manage memory

✓ **Answer: b**

92. **Which scheduling algorithm may lead to starvation?**

a) FCFS

b) Round Robin

c) Priority Scheduling

d) SJF with aging

✓ **Answer: c**

93. **Which of these is a non-preemptive scheduling algorithm?**

a) SJF

b) FCFS

c) Round Robin

d) Multilevel

✓ **Answer: b**

94. **Threads share:**

a) Stack

b) Code

c) Registers

d) PCBs

✓ **Answer: b**

95. **Which is not a reason for context switching?**

- a) I/O request
- b) High-priority process
- c) Memory fragmentation
- d) Timer interrupt

✓ **Answer: c**

96. **Which is the fastest inter-process communication method?**

- a) Message queue
- b) Pipe
- c) Shared memory
- d) Sockets

✓ **Answer: c**

97. **Which model allows threads to be created and managed by the user?**

- a) Kernel-level threads
- b) Microkernel threads
- c) User-level threads
- d) Hybrid threads

✓ **Answer: c**

98. **Thread is also known as:**

- a) Lightweight process
- b) Sub-program
- c) Function
- d) Heavyweight process

✓ **Answer: a**

99. **A thread switching does not require:**

- a) Saving PC
- b) Flushing cache
- c) Saving registers
- d) Changing memory maps

✓ **Answer: d**

100. **Critical section problem is related to:**

- a) Disk scheduling
- b) Memory allocation
- c) Concurrency
- d) Paging

✓ **Answer: c**

✓ **101–150 MCQs on Memory Management**

101. **Which of the following is responsible for memory management?**

- a) Loader
- b) Compiler
- c) Operating System
- d) Linker

✓ **Answer: c**

102. **Which of these is not a memory management technique?**
a) Paging
b) Segmentation
c) Fragmentation
d) Spooling
✓ **Answer: d**
103. **The part of memory that stores the OS and is never paged out is:**
a) Main memory
b) Kernel memory
c) ROM
d) Stack
✓ **Answer: b**
104. **In paging, each process is divided into:**
a) Segments
b) Pages
c) Blocks
d) Frames
✓ **Answer: b**
105. **Which memory allocation scheme suffers from external fragmentation?**
a) Paging
b) Segmentation
c) Contiguous memory allocation
d) Swapping
✓ **Answer: c**
106. **What is the size of a frame in paging?**
a) Variable
b) Depends on process
c) Fixed
d) Depends on memory
✓ **Answer: c**
107. **Page table is used for:**
a) Mapping logical address to physical address
b) Page replacement
c) Disk scheduling
d) Memory compaction
✓ **Answer: a**
108. **The address generated by the CPU is called:**
a) Physical address
b) Logical address
c) Effective address
d) Base address
✓ **Answer: b**
109. **The address seen by the memory unit is:**
a) Logical address
b) Base address
c) Physical address
d) Page number
✓ **Answer: c**

110. **In segmentation, a segment has:**
a) Equal size
b) Fixed size
c) Variable size
d) None
✓ **Answer: c**
111. **What causes internal fragmentation?**
a) Uneven segment sizes
b) Fixed-size memory blocks
c) Variable-size blocks
d) Memory compaction
✓ **Answer: b**
112. **Thrashing occurs when:**
a) All pages are swapped
b) Disk failure happens
c) CPU is idle
d) Too much CPU switching
✓ **Answer: a**
113. **Which is a hardware component that maps virtual to physical address?**
a) MMU
b) ALU
c) CPU
d) RAM
✓ **Answer: a**
114. **Which memory allocation strategy allocates the first hole that is big enough?**
a) Best-fit
b) First-fit
c) Worst-fit
d) Next-fit
✓ **Answer: b**
115. **Which memory allocation strategy leads to the smallest leftover hole?**
a) Best-fit
b) First-fit
c) Worst-fit
d) Equal-fit
✓ **Answer: a**
116. **Segmentation differs from paging because segments are:**
a) Fixed in size
b) Not divided
c) Not used in OS
d) Variable in size
✓ **Answer: d**
117. **Paging avoids which fragmentation?**
a) External
b) Internal
c) Both
d) Neither
✓ **Answer: a**

118. **A page fault occurs when:**
- a) The page is not in memory
 - b) The page is corrupted
 - c) The page is swapped in
 - d) The program crashes
- ✓ **Answer: a**
119. **Which of the following is a page replacement algorithm?**
- a) LRU
 - b) FIFO
 - c) Optimal
 - d) All of the above
- ✓ **Answer: d**
120. **Which page replacement algorithm uses counter to track last use?**
- a) FIFO
 - b) LRU
 - c) Optimal
 - d) Random
- ✓ **Answer: b**
121. **What is the primary purpose of swapping?**
- a) Improve speed
 - b) Improve CPU usage
 - c) Bring in data from secondary storage
 - d) Move process in and out of memory
- ✓ **Answer: d**
122. **Which page replacement algorithm suffers from Belady's Anomaly?**
- a) LRU
 - b) Optimal
 - c) FIFO
 - d) Clock
- ✓ **Answer: c**
123. **What is the purpose of the TLB (Translation Lookaside Buffer)?**
- a) Speed up address translation
 - b) Handle cache miss
 - c) Swap pages
 - d) Control segmentation
- ✓ **Answer: a**
124. **Virtual memory is:**
- a) Main memory
 - b) Memory on disk
 - c) Logical extension of main memory
 - d) None
- ✓ **Answer: c**
125. **Demand paging implies:**
- a) Pages loaded as needed
 - b) All pages loaded at start
 - c) No swapping
 - d) Segmentation
- ✓ **Answer: a**

126. **In a two-level paging system, how many page tables are used?**
a) One
b) Two
c) Depends on size
d) Four
✓ **Answer: b**
127. **Copy-on-write is used for:**
a) Creating new processes
b) Forking with shared memory
c) Paging
d) Disk I/O
✓ **Answer: b**
128. **Memory compaction is:**
a) Eliminating internal fragmentation
b) Removing page faults
c) Shuffling memory contents to reduce external fragmentation
d) Increasing swap space
✓ **Answer: c**
129. **Which of these avoids the need for compaction?**
a) Paging
b) Segmentation
c) Contiguous memory
d) Swapping
✓ **Answer: a**
130. **A TLB miss results in:**
a) Memory fault
b) Page fault
c) Table lookup in page table
d) Process termination
✓ **Answer: c**
131. **The effective memory access time increases due to:**
a) Larger TLB
b) More page faults
c) More RAM
d) Better CPU
✓ **Answer: b**
132. **The minimum number of frames that a process needs is called:**
a) Frame threshold
b) Frame allocation
c) Minimum allocation
d) Degree of multiprogramming
✓ **Answer: c**
133. **Which of the following is used to protect memory?**
a) Base and limit registers
b) Registers
c) Page table
d) TLB
✓ **Answer: a**

134. **Which of these is a memory protection violation?**
a) Page fault
b) Buffer overflow
c) Segmentation fault
d) Stack pointer
✓ **Answer: c**
135. **Which algorithm uses a reference bit for page replacement?**
a) FIFO
b) LRU
c) Clock
d) Optimal
✓ **Answer: c**
136. **What is the page fault rate?**
a) % of time CPU is busy
b) % of processes swapped
c) Ratio of page faults to memory accesses
d) Ratio of segments to pages
✓ **Answer: c**
137. **Which technique allows programs to use more memory than available physically?**
a) Virtual memory
b) Paging
c) Segmentation
d) Swapping
✓ **Answer: a**
138. **Which part of memory is non-swappable?**
a) User process
b) Shell
c) Kernel
d) Data segment
✓ **Answer: c**
139. **The working set model is used to:**
a) Prevent thrashing
b) Determine process priority
c) Replace processes
d) Compact memory
✓ **Answer: a**
140. **What is the cause of thrashing?**
a) Insufficient memory
b) Overuse of CPU
c) Excessive page faults
d) None
✓ **Answer: c**
141. **Which memory region stores environment variables?**
a) Heap
b) Stack
c) Data
d) Code
✓ **Answer: a**

142. **Which memory type is volatile?**
a) ROM
b) Flash
c) RAM
d) Disk
✓ **Answer: c**
143. **What is used for memory address translation?**
a) Page table
b) Index
c) Heap
d) Stack
✓ **Answer: a**
144. **In segmentation, each segment has:**
a) Base and limit
b) Page number
c) Page frame
d) Offset only
✓ **Answer: a**
145. **A page table entry includes:**
a) Page number and frame number
b) Page size
c) CPU ID
d) TLB
✓ **Answer: a**
146. **Segmentation fault usually means:**
a) Memory overflow
b) Invalid memory access
c) Low CPU usage
d) Process killed
✓ **Answer: b**
147. **Page size is typically:**
a) 4 MB
b) 512 KB
c) 4 KB
d) 8 MB
✓ **Answer: c**
148. **Which algorithm is optimal for page replacement?**
a) FIFO
b) Optimal
c) LRU
d) Clock
✓ **Answer: b**
149. **When more processes are loaded than memory can handle, it results in:**
a) Deadlock
b) Thrashing
c) Fragmentation
d) Starvation
✓ **Answer: b**

150. **Which of the following is true for segmentation?**

- a) Each segment has equal size
- b) It avoids both fragmentation types
- c) It allows logical division
- d) Used only in real-time OS

✓ **Answer: c**

✓ **Section D: Deadlocks (151–200)**

151. **What is a deadlock?**

- a) Infinite loop
- b) A process that terminates abnormally
- c) A situation where processes wait indefinitely for resources
- d) A terminated process

✓ **Answer: c)**

152. **Which of the following is NOT a necessary condition for deadlock?**

- a) Mutual exclusion
- b) Hold and wait
- c) Preemption
- d) Circular wait

✓ **Answer: c)**

153. **The circular wait condition for deadlock implies...**

- a) Each process holds one resource
- b) A chain of waiting processes exists
- c) Resources are not reusable
- d) Preemption is allowed

✓ **Answer: b)**

154. **Which of the following can be used to prevent deadlocks?**

- a) Allowing circular wait
- b) Disabling interrupts
- c) Resource ordering
- d) Increasing CPU speed

✓ **Answer: c)**

155. **The deadlock avoidance algorithm that uses a safe state is:**

- a) Peterson's Algorithm
- b) Round Robin
- c) Banker's Algorithm

d) Resource Allocation Graph

✓ **Answer: c)**

156. **What is a safe state in deadlock avoidance?**

- a) A state without any process
- b) A state where all resources are free
- c) A state where the system can allocate resources without leading to deadlock
- d) A state with at most one running process

✓ **Answer: c)**

157. **Which algorithm uses the concept of safe and unsafe states?**

- a) Dijkstra's Algorithm
- b) Banker's Algorithm
- c) Round Robin
- d) FIFO

✓ **Answer: b)**

158. **The resource allocation graph detects deadlock in which scenario?**

- a) Multiple instances
- b) Single instance
- c) Infinite resources
- d) Preemptive OS

✓ **Answer: b)**

159. **Deadlock detection requires:**

- a) Preemptive resources
- b) Safe state checking
- c) A wait-for graph
- d) Multilevel queue scheduling

✓ **Answer: c)**

160. **Which one is a deadlock recovery technique?**

- a) Ignore the deadlock
- b) Resource allocation
- c) Process termination
- d) Multithreading

✓ **Answer: c)**

161. **What is starvation?**
- a) Another name for deadlock
 - b) Process waiting indefinitely due to low priority
 - c) When a process has no resources
 - d) All of the above
- ✓ **Answer: b)**
-

162. **Which strategy ensures the system never enters an unsafe state?**
- a) Deadlock ignorance
 - b) Deadlock detection
 - c) Deadlock prevention
 - d) Deadlock avoidance
- ✓ **Answer: d)**
-

163. **In the Banker's algorithm, the state is safe if:**
- a) Every process gets its maximum need
 - b) No process needs any resource
 - c) The system can allocate resources to all processes in some order
 - d) All processes complete simultaneously
- ✓ **Answer: c)**
-

164. **Which deadlock handling strategy is simplest but risky?**
- a) Prevention
 - b) Avoidance
 - c) Detection and recovery
 - d) Ignorance (Ostrich Algorithm)
- ✓ **Answer: d)**
-

165. **Which of the following is NOT a deadlock recovery method?**
- a) Process termination
 - b) Resource preemption
 - c) Memory swapping
 - d) Process rollback
- ✓ **Answer: c)**
-

166. **What does hold-and-wait mean?**
- a) Holding all resources
 - b) Waiting for CPU
 - c) Holding one resource and waiting for another

d) None of the above

✓ **Answer: c)**

167. **Killing all deadlocked processes is a part of:**

- a) Deadlock prevention
- b) Deadlock detection
- c) Deadlock recovery
- d) Deadlock avoidance

✓ **Answer: c)**

168. **In which case can deadlocks still occur even after detection?**

- a) If resources are not preempted
- b) If processes aren't terminated
- c) If no rollback is done
- d) All of the above

✓ **Answer: d)**

169. **Which graph helps in detecting deadlocks involving single resource instances?**

- a) Resource graph
- b) Wait-for graph
- c) Allocation matrix
- d) Priority tree

✓ **Answer: b)**

170. **What is the key idea behind resource preemption?**

- a) Deny all requests
- b) Stop a process from using resources
- c) Take resource from one process to give another
- d) Remove process permanently

✓ **Answer: c)**

171. **• Which of the following is *not* a necessary condition for deadlock?**

- a) Mutual exclusion
- b) Preemption
- c) Hold and wait
- d) Circular wait

✓ **Answer: b)**

172. **• A deadlock can be prevented by...**

- a) Allowing preemption of resources
- b) Allowing circular wait
- c) Increasing resource count

d) None of the above

✓ **Answer: a)**

173. • **The deadlock avoidance algorithm is based on:**

- a) Safe state
- b) Starvation
- c) Interrupt handling
- d) Context switching

✓ **Answer: a)**

174. • **Which algorithm is used for deadlock avoidance?**

- a) FIFO
- b) SCAN
- c) Banker's
- d) C-SCAN

✓ **Answer: c)**

175. • **A system is in a safe state when:**

- a) There is no deadlock
- b) There is at least one safe sequence
- c) All processes have finished
- d) All processes are waiting

✓ **Answer: b)**

176. • **The Banker's algorithm is used for:**

- a) Memory allocation
- b) Deadlock recovery
- c) CPU scheduling
- d) Deadlock avoidance

✓ **Answer: d)**

177. • **Which condition allows a process to hold resources while waiting for others?**

- a) No preemption
- b) Hold and wait
- c) Circular wait
- d) Mutual exclusion

✓ **Answer: b)**

178. • **Which of the following is the best method to avoid deadlocks in real-time systems?**

- a) Deadlock detection
- b) Deadlock recovery
- c) Deadlock prevention
- d) Deadlock avoidance

✓ **Answer: d)**

179. • **If a system is in an unsafe state:**

- a) Deadlock will occur
- b) Deadlock might occur
- c) Deadlock will not occur
- d) System will shut down

✓ **Answer: b)**

180. • **Deadlock recovery involves:**

- a) Killing all processes

- b) Killing one process at a time
- c) Killing the process with the lowest priority
- d) Killing based on cost and process time

✓ **Answer: d)**

181. • **What does the 'Wait-for graph' detect?**

- a) Safe states
- b) Process scheduling
- c) Deadlocks in single instance resources
- d) Memory leaks

✓ **Answer: c)**

182. • **The method of periodically invoking an algorithm that examines the state of the system to determine whether a deadlock has occurred is called:**

- a) Deadlock prevention
- b) Deadlock detection
- c) Deadlock avoidance
- d) Resource scheduling

✓ **Answer: b)**

183. • **Which of these conditions can be removed to prevent deadlocks?**

- a) Mutual exclusion
- b) Hold and wait
- c) Circular wait
- d) All of the above

✓ **Answer: d)**

184. • **To recover from deadlock, we can:**

- a) Preempt resources
- b) Terminate processes
- c) Roll back processes
- d) All of the above

✓ **Answer: d)**

185. • **A cycle in the resource allocation graph indicates:**

- a) Starvation
- b) Deadlock
- c) Unsafe state
- d) None of the above

✓ **Answer: b)**

186. • **Which statement is true about deadlock detection?**

- a) Detects all types
- b) Cannot be used in multi-resource systems
- c) Works only with safe states
- d) Is used with single instances per resource

✓ **Answer: d)**

187. • **Deadlock avoidance requires that the operating system have advance knowledge of:**

- a) Scheduling time
- b) Resource requirements
- c) Process priorities
- d) I/O patterns

✓ **Answer: b)**

188. • In which state is the system not in deadlock and might avoid it based on future decisions?
- a) Safe
 - b) Unsafe
 - c) Deadlocked
 - d) Idle
- ✓ Answer: b)
189. • Which of the following is a non-recoverable deadlock resolution strategy?
- a) Abort one process
 - b) Rollback
 - c) Request resource again
 - d) Kill all processes
- ✓ Answer: d)
190. • Deadlock recovery techniques include:
- a) Process termination
 - b) Resource preemption
 - c) Process rollback
 - d) All of the above
- ✓ Answer: d)
191. • What is starvation?
- a) Process holding resources
 - b) Process waiting indefinitely
 - c) Process execution completed
 - d) Process blocked
- ✓ Answer: b)
192. • Starvation can be resolved using:
- a) Aging
 - b) Round robin
 - c) Mutual exclusion
 - d) Banker's algorithm
- ✓ Answer: a)
193. • Deadlocks occur mostly in systems with:
- a) Static processes
 - b) Single-threaded processes
 - c) Shared resources
 - d) No memory
- ✓ Answer: c)
194. • Which strategy can cause poor CPU utilization while preventing deadlocks?
- a) Hold and wait
 - b) Circular wait
 - c) Deadlock avoidance
 - d) Deadlock prevention
- ✓ Answer: d)
195. • Which of the following is a necessary condition for deadlock?
- a) All resources preempted
 - b) Processes run in parallel

- c) Resources are shared
- d) Circular wait

✓ **Answer: d)**

196. • **What is the limitation of Banker's algorithm?**

- a) Only works in batch systems
- b) Cannot allocate memory
- c) Requires maximum need in advance
- d) Needs constant preemption

✓ **Answer: c)**

197. • **Which of the following models is used to detect deadlocks?**

- a) Paging model
- b) Spooling model
- c) Wait-for graph model
- d) FAT model

✓ **Answer: c)**

198. • **Which component can lead to a deadlock in a resource allocation graph?**

- a) File descriptors
- b) CPU
- c) I/O
- d) Cycle

✓ **Answer: d)**

199. • **Which strategy increases the priority of a waiting process to resolve starvation?**

- a) Banker's algorithm
- b) Preemptive scheduling
- c) Aging
- d) FCFS

✓ **Answer: c)**

200. • **Deadlocks are more likely to occur in which systems?**

- a) Single-user systems
- b) Time-sharing systems
- c) Multiprocessor systems
- d) Distributed systems

✓ **Answer: d)**

✓ **Section E: File Systems (201–250)**

201. **Which of the following is a valid file system type?**

- a) NTFS
- b) FAT32
- c) ext4
- d) All of the above

✓ **Answer: d)**

202. **The directory structure used in Unix is:**

- a) Linear
- b) Tree
- c) Graph

d) Table

✓ **Answer: b)**

203. Which of the following is not a file attribute?

a) Name

b) Type

c) Length

d) Pipeline

✓ **Answer: d)**

204. Which command in Linux is used to list files?

a) cd

b) ls

c) mkdir

d) pwd

✓ **Answer: b)**

205. File control blocks contain information about:

a) File size

b) Permissions

c) Disk location

d) All of the above

✓ **Answer: d)**

206. Which allocation method causes external fragmentation?

a) Contiguous

b) Linked

c) Indexed

d) Dynamic

✓ **Answer: a)**

207. Which allocation method allows direct access to blocks?

a) Linked

b) Indexed

c) Contiguous

d) Both b and c

✓ **Answer: d)**

208. In a two-level directory system, a user file is found in:

a) Root directory only

b) User directory only

c) System directory only

d) All of the above

✓ **Answer: b)**

209. What does FAT stand for in file systems?

a) File and Access Table

b) File Allocation Table

c) Fast Access Technology

d) File Application Type

✓ **Answer: b)**

210. Which of the following is a disk-based file system?

a) FAT32

b) ext4

c) NTFS

d) All of the above

✓ **Answer: d)**

211. **Which operation is used to create a new file?**

- a) open()
- b) write()
- c) create()
- d) delete()

✓ **Answer: c)**

212. **Which of the following file systems support journaling?**

- a) FAT32
- b) ext3
- c) NTFS
- d) Both b and c

✓ **Answer: d)**

213. **What is the function of a mount operation?**

- a) Deletes a file
- b) Links files together
- c) Makes a file system available
- d) Compresses data

✓ **Answer: c)**

214. **Hard links and symbolic links are related to:**

- a) Memory allocation
- b) File system pointers
- c) Process scheduling
- d) CPU management

✓ **Answer: b)**

215. **Which is faster in access — hard link or symbolic link?**

- a) Hard link
- b) Symbolic link
- c) Both are same
- d) Depends on file type

✓ **Answer: a)**

216. **Which of the following is not true about hard links?**

- a) They refer to the same inode
- b) They work across file systems
- c) They cannot link directories
- d) Deleting one doesn't delete actual data

✓ **Answer: b)**

217. **Which function is used to read data from a file?**

- a) scanf()
- b) write()
- c) read()
- d) getline()

✓ **Answer: c)**

218. **A file pointer in programming refers to:**

- a) File name
- b) File location on disk
- c) Memory address of file stream

d) File size

✓ **Answer: c)**

219. **Which file type is not supported in all OSs?**

a) Executable

b) Pipe

c) FIFO

d) Stream

✓ **Answer: d)**

220. **Sparse files are used to:**

a) Store huge binary files

b) Save disk space for empty blocks

c) Increase access time

d) Protect from corruption

✓ **Answer: b)**

221. **Which directory structure allows sharing of files?**

a) Tree

b) Two-level

c) Acyclic graph

d) Single-level

✓ **Answer: c)**

222. **In Unix, everything is treated as a:**

a) Process

b) File

c) Directory

d) Symbol

✓ **Answer: b)**

223. **The inode in a file system stores:**

a) File name

b) File path

c) Metadata

d) Content

✓ **Answer: c)**

224. **What is true about file descriptors in Unix/Linux?**

a) Negative integers

b) Refer to open files

c) Used only in Windows

d) Store data

✓ **Answer: b)**

225. **What is the default block size in many Linux file systems?**

a) 1 byte

b) 512 bytes

c) 1 KB

d) 4 KB

✓ **Answer: d)**

226. **Which method of file allocation uses index blocks?**

a) Contiguous

b) Linked

c) Indexed

- d) Clustered
✓ **Answer: c)**
227. **What is a directory in OS?**
a) A table of file names and inodes
b) A pointer to memory blocks
c) A type of I/O device
d) None of these
✓ **Answer: a)**
228. **Which command is used to copy files in Unix?**
a) copy
b) cp
c) mv
d) ls
✓ **Answer: b)**
229. **The file opening mode “a+” in C does what?**
a) Reads only
b) Writes only
c) Appends and reads
d) Deletes file content
✓ **Answer: c)**
230. **What is the maximum filename length in many Linux filesystems (ext4)?**
a) 64 bytes
b) 128 bytes
c) 255 bytes
d) Unlimited
✓ **Answer: c)**
231. **Which system call is used to delete a file?**
a) close()
b) fopen()
c) remove()
d) erase()
✓ **Answer: c)**
232. **Sticky bit in Unix is used for:**
a) CPU access control
b) Directory permissions
c) File compression
d) Preventing accidental deletion
✓ **Answer: d)**
233. **What happens when a symbolic link is deleted?**
a) File gets deleted
b) Nothing happens to actual file
c) File becomes corrupted
d) Error occurs
✓ **Answer: b)**
234. **What is a mount point?**
a) Hardware address
b) RAM space
c) Directory to access another file system

d) None

✓ **Answer: c)**

235. **Which file system is used in macOS?**

a) ext3

b) NTFS

c) HFS+ / APFS

d) FAT

✓ **Answer: c)**

236. **Free-space management is needed for:**

a) Memory allocation

b) Disk scheduling

c) File storage

d) Register mapping

✓ **Answer: c)**

237. **Bitmap method is used in file systems for:**

a) CPU scheduling

b) Free space tracking

c) File compression

d) I/O buffering

✓ **Answer: b)**

238. **Which OS file system supports encryption by default?**

a) ext4

b) NTFS

c) FAT

d) UFS

✓ **Answer: b)**

239. **Which of the following file system operations changes file metadata only?**

a) Write

b) chmod

c) read

d) None

✓ **Answer: b)**

240. **What causes internal fragmentation in file systems?**

a) Large file size

b) Mismatched block size and data

c) Inodes

d) File locking

✓ **Answer: b)**

241. **Which directory structure is simplest?**

a) Tree

b) Two-level

c) Single-level

d) DAG

✓ **Answer: c)**

242. **Which of the following is not a valid Linux file type?**

a) Regular file

b) Socket

c) Device

- d) Heap
✓ **Answer: d)**
243. **In which file system is journaling not supported?**
a) ext4
b) ext2
c) NTFS
d) APFS
✓ **Answer: b)**
244. **Which of the following is the smallest unit of allocation in a file system?**
a) Cluster
b) Block
c) Byte
d) Segment
✓ **Answer: b)**
245. **Which of the following allows secure file access over network?**
a) FTP
b) NFS
c) SSHFS
d) telnet
✓ **Answer: c)**
246. **Which command is used to change file permissions in Unix/Linux?**
a) chmod
b) chperm
c) chown
d) ls
✓ **Answer: a)**
247. **Which of the following is true for logical file systems?**
a) Deals with storage blocks
b) Deals with user interface and file naming
c) Manages RAM
d) Controls device I/O
✓ **Answer: b)**
248. **Which of the following is not part of a file system?**
a) Scheduler
b) Metadata
c) File descriptors
d) Directory structure
✓ **Answer: a)**
249. **To remove a directory with files, which command is used in Linux?**
a) rmdir
b) deldir
c) rm -r
d) rm
✓ **Answer: c)**
250. **Which file system is widely used in Android devices?**
a) NTFS
b) FAT
c) ext4

d) APFS

✓ Answer: c)

✓ SECTION F: Disk Scheduling & I/O Management (251–300)

🎯 251. Which disk scheduling algorithm always chooses the request closest to the current head position?

- ☐ a) FCFS
- ☐ b) SSTF
- ☐ c) SCAN
- ☐ d) LOOK

✓ Answer: b) SSTF

🎯 252. Which of the following is *not* a disk scheduling algorithm?

- ☐ a) FCFS
- ☐ b) SCAN
- ☐ c) LIFO
- ☐ d) C-SCAN

✓ Answer: c) LIFO

🎯 253. The SCAN algorithm is sometimes known as:

- ☐ a) Elevator algorithm
- ☐ b) Shuttle algorithm
- ☐ c) Disk crawler
- ☐ d) Spiral scan

✓ Answer: a) Elevator algorithm

🎯 254. What is the primary goal of disk scheduling?

- ☐ a) Minimize memory usage
- ☐ b) Increase CPU usage
- ☐ c) Minimize seek time
- ☐ d) Optimize file structure

✓ Answer: c) Minimize seek time

 **255. Which algorithm moves the head from one end of the disk to the other, servicing requests in one direction only?**

- ☐ a) FCFS
- ☐ b) C-SCAN
- ☐ c) SSTF
- ☐ d) LOOK

☒ **Answer: b) C-SCAN**

 **256. The time taken to position the disk head on the desired track is called:**

- ☐ a) Seek time
- ☐ b) Latency
- ☐ c) Transfer time
- ☐ d) Delay time

☒ **Answer: a) Seek time**

 **257. The time taken to rotate the desired sector under the read/write head is called:**

- ☐ a) Seek time
- ☐ b) Rotational latency
- ☐ c) Transfer time
- ☐ d) Access time

☒ **Answer: b) Rotational latency**

 **258. Which of the following contributes to total disk access time?**

- ☐ a) Seek time
- ☐ b) Rotational latency
- ☐ c) Transfer time
- ☐ d) All of the above

☒ **Answer: d) All of the above**

 **259. Which one provides better performance in case of heavy load?**

- ☐ a) FCFS
 - ☐ b) SSTF
 - ☐ c) SCAN
 - ☐ d) C-SCAN
 - ☒ **Answer: c) SCAN**
-

 **260. Disk scheduling is necessary because:**

- ☐ a) Disks are slower than main memory
 - ☐ b) Disk operations are expensive
 - ☐ c) I/O requests may be numerous and scattered
 - ☐ d) All of the above
 - ☒ **Answer: d) All of the above**
-

 **261. Which of the following is a drawback of SSTF?**

- ☐ a) Causes starvation
 - ☐ b) Complex algorithm
 - ☐ c) Ignores disk geometry
 - ☐ d) High memory usage
 - ☒ **Answer: a) Causes starvation**
-

 **262. Which scheduling algorithm can lead to starvation of some requests?**

- ☐ a) SCAN
 - ☐ b) SSTF
 - ☐ c) FCFS
 - ☐ d) C-SCAN
 - ☒ **Answer: b) SSTF**
-

 **263. LOOK scheduling differs from SCAN in that it:**

- ☐ a) Ignores queue
- ☐ b) Doesn't go to the end of disk
- ☐ c) Uses priority
- ☐ d) Works like SSTF
- ☒ **Answer: b) Doesn't go to the end of disk**

 **264. In C-LOOK, the head moves:**

- ☐ a) Back and forth
- ☐ b) From center outward
- ☐ c) In one direction only and jumps back
- ☐ d) Randomly
- ☒ **Answer: c) In one direction only and jumps back**

 **265. The total time to access data on disk is:**

- ☐ a) Transfer time
- ☐ b) Seek time + latency
- ☐ c) Transfer + seek + latency
- ☐ d) Seek + latency - transfer
- ☒ **Answer: c) Transfer + seek + latency**

 **266. What is disk bandwidth?**

- ☐ a) Bits per track
- ☐ b) Tracks per cylinder
- ☐ c) Data transfer rate
- ☐ d) Seek speed
- ☒ **Answer: c) Data transfer rate**

 **267. What causes disk fragmentation?**

- ☐ a) Full memory
- ☐ b) Data compression
- ☐ c) Files stored non-contiguously
- ☐ d) File overwrites
- ☒ **Answer: c) Files stored non-contiguously**

 **268. What does defragmentation do?**

- ☐ a) Deletes old files
 - ☐ b) Removes duplicate files
 - ☐ c) Rearranges files contiguously
 - ☐ d) Encrypts files
 - ☒ **Answer: c) Rearranges files contiguously**
-

 **269. Which of these is true for FCFS disk scheduling?**

- ☐ a) High throughput
 - ☐ b) No starvation
 - ☐ c) High performance
 - ☐ d) Low fairness
 - ☒ **Answer: b) No starvation**
-

 **270. The best disk scheduling algorithm for fairness is:**

- ☐ a) SSTF
 - ☐ b) SCAN
 - ☐ c) FCFS
 - ☐ d) LOOK
 - ☒ **Answer: c) FCFS**
-

 **271. If multiple I/O requests are pending, which algorithm gives minimum average response time?**

- ☐ a) FCFS
 - ☐ b) SSTF
 - ☐ c) SCAN
 - ☐ d) C-LOOK
 - ☒ **Answer: b) SSTF**
-

 **272. Seek time depends on:**

- ☐ a) Track layout
- ☐ b) Disk RPM
- ☐ c) Head movement

- ☐ d) File size
☒ **Answer: c) Head movement**
-

 **273. Rotational latency depends on:**

- ☐ a) Size of disk
☐ b) Disk RPM
☐ c) File type
☐ d) Cache speed
☒ **Answer: b) Disk RPM**
-

 **274. Transfer time is affected by:**

- ☐ a) Size of data
☐ b) Seek time
☐ c) Latency
☐ d) OS
☒ **Answer: a) Size of data**
-

 **275. Which scheduling technique uses a queue that's circular in nature?**

- ☐ a) SCAN
☐ b) C-SCAN
☐ c) SSTF
☐ d) FCFS
☒ **Answer: b) C-SCAN**
-

 **275. Which scheduling algorithm may cause starvation?**

- ☐ a) FCFS
☐ b) SSTF
☐ c) SCAN
☐ d) C-SCAN
☒ **Answer: b) SSTF**
-

 **276. LOOK scheduling algorithm differs from SCAN in that:**

- ☐ a) It doesn't reverse direction
 - ☐ b) It goes to end of the disk
 - ☐ c) It only scans middle cylinders
 - ☐ d) It reverses without going to the end
 - ☒ **Answer: d) It reverses without going to the end**
-

 **277. Which disk scheduling technique avoids unnecessary movement of the disk arm?**

- ☐ a) FCFS
 - ☐ b) SSTF
 - ☐ c) LOOK
 - ☐ d) C-SCAN
 - ☒ **Answer: c) LOOK**
-

 **278. The average seek time is lowest for which of the following?**

- ☐ a) FCFS
 - ☐ b) SCAN
 - ☐ c) SSTF
 - ☐ d) C-SCAN
 - ☒ **Answer: c) SSTF**
-

 **279. C-LOOK differs from C-SCAN in that it:**

- ☐ a) Doesn't service requests
 - ☐ b) Goes to the end before reversing
 - ☐ c) Goes to last request before reversing
 - ☐ d) Is not a valid algorithm
 - ☒ **Answer: c) Goes to last request before reversing**
-

 **280. What does FCFS stand for in disk scheduling?**

- ☐ a) First Come First Served
- ☐ b) First Compute First Schedule
- ☐ c) Fast Cylinder Fast Schedule

- ☐ d) First Completion First Selection
☒ **Answer: a) First Come First Served**
-

 **281. In SCAN disk scheduling, the head:**

- ☐ a) Moves back and forth endlessly
☐ b) Moves only in one direction
☐ c) Moves to center and returns
☐ d) Moves to end and reverses
☒ **Answer: d) Moves to end and reverses**
-

 **282. Which of the following is best for a system with heavy disk load?**

- ☐ a) FCFS
☐ b) SSTF
☐ c) SCAN
☐ d) C-SCAN
☒ **Answer: c) SCAN**
-

 **283. Disk I/O performance can be improved by:**

- ☐ a) Better scheduling
☐ b) Increasing CPU speed
☐ c) Decreasing main memory
☐ d) Adding more cables
☒ **Answer: a) Better scheduling**
-

 **284. Disk latency is the sum of:**

- ☐ a) Seek time and transfer time
☐ b) Seek time and rotational latency
☐ c) Transfer time and CPU time
☐ d) None of these
☒ **Answer: b) Seek time and rotational latency**
-

 **285. In real-time systems, which disk scheduling is preferred?**

- ☐ a) SSTF
- ☐ b) FCFS
- ☐ c) C-SCAN
- ☐ d) SCAN

☒ **Answer: b) FCFS**

 **286. Disk blocks are grouped into:**

- ☐ a) Clusters
- ☐ b) Buffers
- ☐ c) Frames
- ☐ d) Pages

☒ **Answer: a) Clusters**

 **287. The controller that handles disk operations is called:**

- ☐ a) CPU
- ☐ b) DMA
- ☐ c) Device driver
- ☐ d) Disk controller

☒ **Answer: d) Disk controller**

 **288. The unit of data transfer between disk and memory is:**

- ☐ a) Byte
- ☐ b) Sector
- ☐ c) Word
- ☐ d) Page

☒ **Answer: b) Sector**

 **289. Which mechanism enhances disk access speed?**

- ☐ a) Disk caching
- ☐ b) RAID
- ☐ c) SSD

- ☐ d) All of the above
- ✓ **Answer: d) All of the above**
-

🎯 **290. Rotational latency depends on:**

- ☐ a) Disk size
- ☐ b) Head size
- ☐ c) Disk speed
- ☐ d) File size
- ✓ **Answer: c) Disk speed**
-

🎯 **291. Disk striping is used to:**

- ☐ a) Decrease capacity
- ☐ b) Increase seek time
- ☐ c) Improve throughput
- ☐ d) Protect metadata
- ✓ **Answer: c) Improve throughput**
-

🎯 **292. What is RAID?**

- ☐ a) Random Access and Input Device
- ☐ b) Redundant Array of Independent Disks
- ☐ c) Repetitive Array of Identical Drives
- ☐ d) Remote Access Internet Device
- ✓ **Answer: b) Redundant Array of Independent Disks**
-

🎯 **293. RAID improves:**

- ☐ a) Storage speed
- ☐ b) Fault tolerance
- ☐ c) Both a and b
- ☐ d) Only security
- ✓ **Answer: c) Both a and b**
-

 **294. The fastest RAID level for read operations is:**

- ☐ a) RAID 0
- ☐ b) RAID 1
- ☐ c) RAID 5
- ☐ d) RAID 10

☒ **Answer: a) RAID 0**

 **295. RAID 1 is also known as:**

- ☐ a) Striping
- ☐ b) Mirroring
- ☐ c) Parity
- ☐ d) Cloning

☒ **Answer: b) Mirroring**

 **296. RAID 5 provides:**

- ☐ a) Mirroring
- ☐ b) No redundancy
- ☐ c) Parity-based redundancy
- ☐ d) Hot-swapping only

☒ **Answer: c) Parity-based redundancy**

 **297. In disk organization, a sector is:**

- ☐ a) A block of memory
- ☐ b) A part of a page
- ☐ c) A segment of a disk track
- ☐ d) A disk controller

☒ **Answer: c) A segment of a disk track**

 **298. The controller stores and manages:**

- ☐ a) Only data
- ☐ b) I/O queue
- ☐ c) File system

- ☐ d) CPU registers
☒ **Answer: b) I/O queue**
-

 **299. Transfer rate is defined as:**

- ☐ a) Data per second from memory to CPU
☐ b) Data per second from disk to memory
☐ c) Bits stored on CPU
☐ d) Bytes fetched from cache
☒ **Answer: b) Data per second from disk to memory**
-

 **300. In disk scheduling, starvation is caused due to:**

- ☐ a) Long waiting queue
☐ b) Prioritizing close requests
☐ c) Rotational delay
☐ d) Disk failure
☒ **Answer: b) Prioritizing close requests**
-

SECTION G: File System & Storage Management (301 – 350)

 **301. The file system manages which of the following?**

- ☐ a) Files
☐ b) Directories
☐ c) Disk space
☐ d) All of the above
☒ **Answer: d) All of the above**
-

 **302. Which allocation method uses pointers to connect file blocks?**

- ☐ a) Contiguous
☐ b) Indexed
☐ c) Linked
☐ d) Direct
☒ **Answer: c) Linked**
-

 **303. The directory structure used in UNIX is:**

- ☐ a) Single-level
 - ☐ b) Two-level
 - ☐ c) Tree-structured
 - ☐ d) Acyclic
- ☒ **Answer: c) Tree-structured**
-

 **304. The FAT (File Allocation Table) is used in which operating system?**

- ☐ a) UNIX
 - ☐ b) Linux
 - ☐ c) MS-DOS
 - ☐ d) Mac OS
- ☒ **Answer: c) MS-DOS**
-

 **305. A file control block (FCB) contains information about:**

- ☐ a) File size
 - ☐ b) File permissions
 - ☐ c) File location
 - ☐ d) All of the above
- ☒ **Answer: d) All of the above**
-

 **306. Which of the following is *not* a file access method?**

- ☐ a) Sequential
 - ☐ b) Indexed
 - ☐ c) Random
 - ☐ d) Hashed
- ☒ **Answer: d) Hashed**
-

 **307. What is the purpose of a directory in a file system?**

- ☐ a) Store data
- ☐ b) Store metadata
- ☐ c) Map filenames to file control blocks

- ☐ d) Handle I/O
- ✓ **Answer: c) Map filenames to file control blocks**
-

🔗 **308. Which file organization is the most efficient in terms of space?**

- ☐ a) Contiguous
- ☐ b) Linked
- ☐ c) Indexed
- ☐ d) Hashed
- ✓ **Answer: a) Contiguous**
-

🔗 **309. In indexed allocation, the index block contains:**

- ☐ a) Actual file data
- ☐ b) File names
- ☐ c) Pointers to data blocks
- ☐ d) File attributes
- ✓ **Answer: c) Pointers to data blocks**
-

🔗 **310. Which of the following is true about hard links in UNIX?**

- ☐ a) They link to the file name
- ☐ b) They are symbolic shortcuts
- ☐ c) They are different from the original file
- ☐ d) They share the same inode
- ✓ **Answer: d) They share the same inode**

✓ **311. Which allocation method allows files to grow dynamically and minimizes disk fragmentation?**

- a) Contiguous Allocation
- b) Linked Allocation
- c) Indexed Allocation
- d) Multilevel Allocation
- ✓ **Answer: b) Linked Allocation**
-

✓ **312. In indexed allocation, the index block contains:**

- a) Actual file data
- b) Pointers to data blocks
- c) Directory entries
- d) File names

✓ Answer: b) Pointers to data blocks

✓ 313. Which disk scheduling algorithm is also known as the elevator algorithm?

- a) FCFS
- b) SSTF
- c) SCAN
- d) C-SCAN

✓ Answer: c) SCAN

✓ 314. The C-SCAN algorithm:

- a) Services requests in one direction only
- b) Is the same as SCAN
- c) Causes starvation
- d) Services requests circularly

✓ Answer: d) Services requests circularly

✓ 315. The time taken to position the read/write head over the correct track is called:

- a) Seek time
- b) Latency time
- c) Transfer time
- d) Access time

✓ Answer: a) Seek time

✓ 316. Which is NOT a file access method?

- a) Sequential
- b) Direct
- c) Random
- d) Indexed

✓ Answer: d) Indexed

✓ **317. What is the role of the file control block (FCB)?**

- a) Stores file content
- b) Keeps file metadata
- c) Allocates memory
- d) Manages I/O devices

✓ **Answer: b) Keeps file metadata**

✓ **318. Which allocation method is best for small files?**

- a) Contiguous
- b) Linked
- c) Indexed
- d) Hybrid

✓ **Answer: a) Contiguous**

✓ **319. Which of the following is a disk scheduling algorithm?**

- a) FIFO
- b) LRU
- c) SSTF
- d) FCFS

✓ **Answer: c) SSTF**

✓ **320. Which strategy improves performance by buffering disk writes?**

- a) Disk mirroring
- b) RAID
- c) Write-back caching
- d) Sequential scheduling

✓ **Answer: c) Write-back caching**

✓ **321. FAT (File Allocation Table) is used in:**

- a) UNIX
- b) NTFS
- c) Windows (older versions)

d) MacOS

✓ **Answer: c) Windows (older versions)**

✓ **322. Which RAID level uses disk striping with parity?**

a) RAID 0

b) RAID 1

c) RAID 5

d) RAID 10

✓ **Answer: c) RAID 5**

✓ **323. In disk scheduling, SSTF stands for:**

a) Shortest Seek Time First

b) Shortest Scheduled Time First

c) Synchronous Scheduling Technique First

d) Sequential Seek Track First

✓ **Answer: a) Shortest Seek Time First**

✓ **324. Which system call is used to delete a file?**

a) open()

b) read()

c) write()

d) unlink()

✓ **Answer: d) unlink()**

✓ **325. Which component manages disk quotas and limits?**

a) Process Manager

b) File System

c) Kernel

d) Shell

✓ **Answer: b) File System**

✓ **326. What causes disk fragmentation?**

- a) Contiguous allocation
 - b) Deleting and creating files over time
 - c) Indexed allocation
 - d) Using SSDs
- ✓ **Answer: b) Deleting and creating files over time**
-

✓ **327. A UNIX inode stores:**

- a) Only file content
 - b) File metadata like size, permissions
 - c) User information
 - d) Only file names
- ✓ **Answer: b) File metadata like size, permissions**
-

✓ **328. Disk bandwidth is defined as:**

- a) The number of bytes transferred per second
 - b) The speed of the disk
 - c) The time taken to read/write
 - d) Number of files on the disk
- ✓ **Answer: a) The number of bytes transferred per second**
-

✓ **329. Which disk scheduling algorithm leads to starvation?**

- a) FCFS
 - b) SSTF
 - c) SCAN
 - d) LOOK
- ✓ **Answer: b) SSTF**
-

✓ **330. In UNIX, the `mount` command is used to:**

- a) Format a disk
 - b) Allocate memory
 - c) Attach a file system
 - d) View file permissions
- ✓ **Answer: c) Attach a file system**
-

✓ **331. What does the term “thrashing” relate to in context of file systems?**

- a) File deletion
- b) Page faults
- c) Repeated disk access
- d) Disk I/O overflow

✓ **Answer: c) Repeated disk access**

✓ **332. Which file allocation method uses an index block?**

- a) Contiguous
- b) Linked
- c) Indexed
- d) Clustered

✓ **Answer: c) Indexed**

✓ **333. Directory structure where users have separate subdirectories is:**

- a) Single-level
- b) Two-level
- c) Tree-structured
- d) Acyclic

✓ **Answer: b) Two-level**

✓ **334. Which command shows file system disk usage in UNIX?**

- a) ls
- b) du
- c) df
- d) ps

✓ **Answer: c) df**

✓ **335. What is the function of a boot block?**

- a) Contains OS drivers
- b) Stores BIOS info
- c) Loads the OS
- d) Contains user files

✓ **Answer: c) Loads the OS**

✓ **336. Which is not a file attribute?**

- a) Name
- b) Date
- c) Size
- d) Owner's age

✓ **Answer: d) Owner's age**

✓ **337. In file systems, ACL stands for:**

- a) Access Control List
- b) Auto Control Loop
- c) Active Command Line
- d) Allocation Control List

✓ **Answer: a) Access Control List**

✓ **338. What causes internal fragmentation in file systems?**

- a) Unused space within blocks
- b) Too many files
- c) Repeated writes
- d) Lack of memory

✓ **Answer: a) Unused space within blocks**

✓ **339. Which RAID level mirrors data across two disks?**

- a) RAID 0
- b) RAID 1
- c) RAID 5
- d) RAID 10

✓ **Answer: b) RAID 1**

✓ **340. A file's logical structure is defined by:**

- a) Operating System
- b) File format
- c) File header

d) Application program

✓ **Answer: d) Application program**

✓ **341. In UNIX, soft link is also called:**

- a) Inode link
- b) Symbolic link
- c) Hard link
- d) Block link

✓ **Answer: b) Symbolic link**

✓ **342. Which disk scheduling technique is most fair but least efficient?**

- a) SCAN
- b) FCFS
- c) SSTF
- d) LOOK

✓ **Answer: b) FCFS**

✓ **343. A file system must provide:**

- a) Encryption only
- b) Storage only
- c) Backup only
- d) Data access, protection, and recovery

✓ **Answer: d) Data access, protection, and recovery**

✓ **344. Which structure contains directory info in UNIX?**

- a) FAT
- b) inode
- c) Superblock
- d) Mount table

✓ **Answer: c) Superblock**

✓ **345. External fragmentation occurs when:**

- a) Blocks are too small
- b) Free memory is scattered
- c) Disk crashes
- d) Files are deleted

✓ Answer: b) Free memory is scattered

✓ 346. Which file system is used by most Linux distributions?

- a) FAT32
- b) ext4
- c) NTFS
- d) HFS

✓ Answer: b) ext4

✓ 347. Which of the following allows hot-swapping in storage?

- a) HDD
- b) RAID
- c) SSD
- d) NAS

✓ Answer: d) NAS

✓ 348. Which of the following RAID levels offers both redundancy and performance?

- a) RAID 0
- b) RAID 1
- c) RAID 5
- d) RAID 10

✓ Answer: d) RAID 10

✓ 349. In file allocation, FAT maintains:

- a) File headers
- b) Cluster chains
- c) Bitmaps
- d) Inodes

✓ Answer: b) Cluster chains

✅ **350. Which of the following is true about journaling file systems?**

- a) They are slower
- b) They record metadata changes
- c) They reduce security
- d) They prevent indexing

✔ **Answer: b) They record metadata changes**

📁 🖥️ **SECTION H: System Calls, Deadlocks & Scheduling (351 – 400)** 🖥️ 📁

🔖 *Advanced MCQs covering process synchronization, scheduling algorithms, system calls, and deadlock management.*

351. What is the purpose of a system call in an operating system?

- A) To provide an interface between user and hardware
- B) To manage I/O devices
- C) To allow user-level processes to request services from the OS
- D) To allow direct access to hardware

✅ **Answer: C**

352. A system call used to create a new process is:

- A) fork()
- B) exit()
- C) wait()
- D) exec()

✅ **Answer: A**

353. Which system call is used to replace a process image?

- A) wait()
- B) fork()
- C) exec()
- D) create()

✅ **Answer: C**

354. What is the main problem of the Fork-Join model?

- A) Process creation
- B) Deadlocks
- C) Process synchronization
- D) Resource sharing

✅ **Answer: C**

355. A process is in the “blocked” state when:

- A) It is executing
- B) It is waiting for I/O

- C) It has finished execution
- D) It is ready to run

✓ **Answer: B**

356. What is a **deadlock**?

- A) Infinite loop
- B) Infinite blocking between processes
- C) An error in code
- D) System crash

✓ **Answer: B**

357. Which of the following is **NOT** a necessary condition for deadlock?

- A) Mutual exclusion
- B) Hold and wait
- C) Preemption
- D) Circular wait

✓ **Answer: C**

358. What is **deadlock prevention**?

- A) Ignoring the problem
- B) Avoiding circular wait
- C) Killing all processes
- D) Restarting OS

✓ **Answer: B**

359. Banker's algorithm is used for:

- A) File allocation
- B) Deadlock detection
- C) Deadlock avoidance
- D) Disk scheduling

✓ **Answer: C**

360. In Banker's algorithm, the system is in a safe state if:

- A) All processes are blocked
- B) Deadlock occurs
- C) There exists a safe sequence
- D) Processes run concurrently

✓ **Answer: C**

361. Which is NOT a deadlock handling method?

- A) Deadlock prevention
- B) Deadlock detection and recovery
- C) Deadlock ignoring
- D) Deadlock renaming

✓ **Answer: D**

362. CPU scheduling deals with:

- A) Disk management
- B) Allocation of CPU to processes

- C) File access
- D) Page replacement

✓ **Answer: B**

363. Which CPU scheduling algorithm may lead to **starvation**?

- A) FCFS
- B) Round Robin
- C) SJF
- D) Priority Scheduling

✓ **Answer: D**

364. In which algorithm is the **turnaround time** minimized?

- A) FCFS
- B) SJF
- C) Round Robin
- D) Priority

✓ **Answer: B**

365. Round Robin scheduling is most suitable for:

- A) Real-time systems
- B) Batch systems
- C) Time-sharing systems
- D) Embedded systems

✓ **Answer: C**

366. Which is preemptive scheduling?

- A) SJF
- B) FCFS
- C) Priority
- D) Round Robin

✓ **Answer: D**

367. Time quantum in Round Robin affects:

- A) Context switch rate
- B) CPU usage
- C) Throughput
- D) Deadlock

✓ **Answer: A**

368. What is context switching?

- A) Swapping two processes
- B) Switching from user mode to kernel
- C) Saving state of a process and loading another
- D) Switching from kernel to user mode

✓ **Answer: C**

369. Aging technique is used to:

- A) Increase CPU time
- B) Avoid starvation

- C) Increase waiting time
- D) Prevent paging

✓ **Answer: B**

370. The shortest job next (SJN) scheduling is:

- A) Preemptive
- B) Non-preemptive
- C) Both
- D) None

✓ **Answer: B**

371. In **Multilevel Queue Scheduling**, processes are:

- A) All handled equally
- B) Grouped by priority or type
- C) Scheduled only by FCFS
- D) Killed when idle

✓ **Answer: B**

372. Which of the following gives maximum CPU utilization?

- A) Round Robin
- B) SJF
- C) FCFS
- D) Priority Scheduling

✓ **Answer: B**

373. A starvation-free scheduling algorithm is:

- A) FCFS
- B) SJF
- C) Priority
- D) Multilevel Queue

✓ **Answer: A**

374. Dispatcher is used to:

- A) Select process
- B) Execute process
- C) Assign CPU
- D) Context switch

✓ **Answer: D**

375. Preemptive scheduling can result in:

- A) Reduced overhead
- B) More context switches
- C) Improved priority
- D) Static scheduling

✓ **Answer: B**

376. What is a process state diagram?

- A) Code map
- B) Thread map

- C) Visual of process transitions
- D) Scheduling tree

✓ **Answer: C**

377. Which is NOT a state in process life cycle?

- A) Ready
- B) Running
- C) Waiting
- D) Terminating

✓ **Answer: D**

378. In multithreading, threads share:

- A) Registers
- B) Stack
- C) Code section
- D) Program counter

✓ **Answer: C**

379. The degree of multiprogramming is:

- A) Number of CPU cores
- B) Number of processes in memory
- C) Number of I/O devices
- D) Number of system calls

✓ **Answer: B**

380. What does the **wait()** system call do?

- A) Makes process sleep
- B) Terminates the process
- C) Waits for a child process to finish
- D) Checks I/O

✓ **Answer: C**

381. Which is used for inter-process communication?

- A) `exec()`
- B) `fork()`
- C) `pipe()`
- D) `exit()`

✓ **Answer: C**

382. What does **exec()** do in process creation?

- A) Terminates parent
- B) Replaces process image
- C) Starts multithreading
- D) Forks new process

✓ **Answer: B**

383. What happens if `exec()` is successful?

- A) It returns to the parent
- B) The original program continues

- C) The original program is replaced
- D) It forks another child

✓ **Answer: C**

384. Mutual exclusion ensures:

- A) Shared use
- B) Exclusive access to critical section
- C) Multiple access to data
- D) Program termination

✓ **Answer: B**

385. Race condition occurs when:

- A) Multiple threads wait
- B) Multiple threads update shared data
- C) Only one process runs
- D) I/O completes

✓ **Answer: B**

386. Which scheduling algorithm is ideal for **interactive systems**?

- A) FCFS
- B) SJF
- C) Round Robin
- D) Priority

✓ **Answer: C**

387. What is the **safe sequence** in Banker's Algorithm?

- A) Process ID
- B) Sequence with no deadlock
- C) Maximum claim
- D) Turnaround order

✓ **Answer: B**

388. The degree of concurrency in OS is:

- A) Number of running processes
- B) Number of logical processors
- C) Number of concurrent threads
- D) Number of system calls

✓ **Answer: C**

389. A semaphore is:

- A) A process
- B) A variable used to control access
- C) A system call
- D) A memory segment

✓ **Answer: B**

390. A binary semaphore takes values:

- A) 0 and 1
- B) 0 to 10

- C) -1 and 1
- D) Boolean
- ✓ **Answer: A**

391. The critical section problem involves:

- A) Hardware
- B) Memory
- C) Shared resources
- D) Threads

✓ **Answer: C**

392. Monitor is used in synchronization to:

- A) Monitor CPU
- B) Synchronize processes
- C) Detect deadlock
- D) Block threads

✓ **Answer: B**

393. A spinlock is used to:

- A) Delay process
- B) Busy wait for lock
- C) Avoid context switch
- D) Implement FCFS

✓ **Answer: B**

394. Peterson's solution is for:

- A) Process scheduling
- B) Memory allocation
- C) Critical section problem
- D) Thread synchronization

✓ **Answer: C**

395. Which of the following is a blocking synchronization mechanism?

- A) Busy waiting
- B) Spinlocks
- C) Semaphores
- D) Interrupts

✓ **Answer: C**

396. Starvation can be prevented by:

- A) Deadlock
- B) Aging
- C) Scheduling
- D) Monitors

✓ **Answer: B**

397. What is the **main issue** with non-preemptive scheduling?

- A) Starvation
- B) Low throughput

- C) Long wait time for high-priority tasks
- D) High context switch

✓ **Answer: C**

398. In a non-preemptive system, CPU is:

- A) Taken anytime
- B) Taken by interrupt
- C) Given only when process finishes
- D) Given randomly

✓ **Answer: C**

399. What does **thrashing** relate to?

- A) Process scheduling
- B) Excessive context switching
- C) Memory paging
- D) File swapping

✓ **Answer: C**

400. Deadlock can occur when processes:

- A) Share resources non-exclusively
- B) Access all resources at once
- C) Request and hold resources
- D) Never request I/O

✓ **Answer: C**

SECTION I: Virtual Memory, Paging, and Segmentation (401–450)

 *Master concepts like page tables, segmentation, TLB, fragmentation, and address translation.*

401. Virtual memory is:

- A) Part of physical memory
- B) Used only in Linux
- C) Memory on the hard disk
- D) An illusion of a large main memory

✓ **Answer: D**

402. The purpose of virtual memory is to:

- A) Increase CPU speed
- B) Allow more programs to run than physical memory

- C) Eliminate I/O
- D) Make cache memory

✓ Answer: B

403. Paging helps in:

- A) Memory allocation
- B) CPU scheduling
- C) Disk formatting
- D) Process synchronization

✓ Answer: A

404. Page table stores:

- A) Disk addresses
- B) Memory contents
- C) Mapping of virtual to physical addresses
- D) Register contents

✓ Answer: C

405. What causes a page fault?

- A) Hardware error
- B) Memory full
- C) Page not in main memory
- D) Segment missing

✓ Answer: C

406. In paging, physical memory is divided into:

- A) Pages
- B) Segments
- C) Frames
- D) Blocks

✓ Answer: C

407. Which technique reduces page fault rate?

- A) Caching
- B) Thrashing
- C) Working set model
- D) TLB flushing

✓ Answer: C

408. TLB stands for:

- A) Temporary Load Buffer
- B) Translation Lookaside Buffer
- C) Translated Logical Bit
- D) Transfer Level Buffer

✓ Answer: B

409. What is stored in the TLB?

- A) CPU registers
- B) Cache lines
- C) Page table entries
- D) Disk blocks

✓ Answer: C

410. Segmentation allows:

- A) Equal memory division
- B) Variable-size memory allocation
- C) Fixed-size paging
- D) Memory leaks

✓ Answer: B

411. Thrashing occurs when:

- A) Disk crashes
- B) Too many processes run
- C) Too many page faults occur
- D) Memory is overloaded

✓ Answer: C

412. Effective access time depends on:

- A) CPU clock
- B) TLB hit ratio
- C) I/O speed
- D) File size

☒ **Answer: B**

413. Page size is defined by:

- A) Operating system
- B) Hardware
- C) User
- D) BIOS

☒ **Answer: B**

414. The valid-invalid bit is used in:

- A) Registers
- B) Page replacement
- C) Page table
- D) Cache

☒ **Answer: C**

415. Which of the following suffers from external fragmentation?

- A) Paging
- B) Segmentation
- C) Combined Paging-Segmentation
- D) Virtual memory

☒ **Answer: B**

416. Which suffers from internal fragmentation?

- A) Segmentation
- B) Paging
- C) Virtual memory
- D) Demand paging

☒ **Answer: B**

417. Which memory management scheme uses variable-sized blocks?

- A) Paging
- B) Segmentation
- C) Cache memory
- D) TLB

✓ **Answer: B**

418. The basic unit of data transfer between disk and memory in paging is:

- A) Page
- B) Frame
- C) Segment
- D) Block

✓ **Answer: A**

419. Demand paging loads pages:

- A) All at once
- B) When needed
- C) At compile time
- D) Never

✓ **Answer: B**

420. In a system using paging, the address generated by the CPU is known as:

- A) Physical address
- B) Logical address
- C) Frame address
- D) Virtual segment

✓ **Answer: B**

 **421. What is a page fault?**

- A) A page that is successfully loaded into memory
- B) A page not present in the memory when referenced
- C) A disk failure
- D) A corrupted memory block

Ans: B

 **422. What is the purpose of the Translation Lookaside Buffer (TLB)?**

- A) Store frequently used data
- B) Increase CPU speed
- C) Reduce address translation time
- D) Manage cache memory

Ans: C

 **423. What is the main advantage of paging over segmentation?**

- A) Pages are variable sized
- B) Avoids external fragmentation
- C) Easier to manage variable data
- D) Increases memory usage

Ans: B

 **424. In demand paging, when is a page loaded into memory?**

- A) At program startup
- B) When the system boots
- C) Only when it is required during execution
- D) When all pages are required

Ans: C

 **425. Which of the following can cause thrashing?**

- A) Too many pages in memory
- B) Large CPU cache
- C) High CPU utilization
- D) High page fault rate

Ans: D

 **426. What is the function of the page table?**

- A) Stores the actual data
- B) Maps logical addresses to physical addresses
- C) Stores program counters

D) Holds cache data

Ans: B

 **427. Which technique allows more processes to be loaded than physical memory?**

- A) Segmentation
- B) Multitasking
- C) Virtual Memory
- D) Real-time scheduling

Ans: C

 **428. What is the size of a page in most modern systems?**

- A) 128 Bytes
- B) 4 KB
- C) 512 KB
- D) 1 MB

Ans: B

 **429. What is internal fragmentation?**

- A) Unused memory between partitions
- B) Lost memory within allocated space
- C) Lost memory outside memory
- D) None of the above

Ans: B

 **430. Which data structure is essential for virtual memory systems?**

- A) Page Table
- B) Disk Scheduler
- C) CPU Scheduler
- D) Thread Pool

Ans: A

 **431. What is a frame in paging?**

- A) A virtual address
- B) A memory block in physical memory
- C) A memory block in logical memory
- D) A disk block

Ans: B

 **432. The effective memory access time in a paging system depends on:**

- A) Page size
- B) TLB hit ratio
- C) Number of pages
- D) Segments

Ans: B

 **433. The replacement of a page in memory is done using:**

- A) Page Table
- B) Page Replacement Algorithm
- C) Memory Allocator
- D) Stack

Ans: B

 **434. Which page replacement algorithm uses the oldest page?**

- A) FIFO
- B) LRU
- C) Optimal
- D) Clock

Ans: A

 **435. The Optimal Page Replacement algorithm replaces:**

- A) Most frequently used page
- B) Least recently used page
- C) Page that will not be used for longest time
- D) Random page

Ans: C

 **436. Which is the major disadvantage of the Optimal algorithm?**

- A) Too complex
- B) Requires future knowledge of requests
- C) Slow
- D) Doesn't work on large data

Ans: B

 **437. LRU stands for:**

- A) Last Recently Used
- B) Least Random Usage
- C) Least Recently Used
- D) Latest Read Unit

Ans: C

 **438. Which of the following is not a page replacement algorithm?**

- A) FIFO
- B) Optimal
- C) LRU
- D) Round Robin

Ans: D

 **439. In segmentation, each segment has:**

- A) Fixed size
- B) Variable size
- C) Same address
- D) Only one value

Ans: B

 **440. Segmentation is visible to the:**

- A) Operating System only
- B) User
- C) Hardware
- D) Scheduler

Ans: B

 **441. Paging is a form of:**

- A) Contiguous allocation
- B) Fragmentation
- C) Non-contiguous allocation
- D) Static partitioning

Ans: C

 **442. What causes internal fragmentation?**

- A) Variable-sized partitions
- B) Fixed-sized partitions or pages
- C) Memory holes
- D) Page faults

Ans: B

 **443. What causes external fragmentation?**

- A) Fixed-sized allocation
- B) Memory leaks
- C) Variable-sized memory blocks
- D) Stack overflow

Ans: C

 **444. A page table entry contains:**

- A) Logical address only
- B) Physical address only
- C) Both logical and physical address
- D) Frame number and control bits

Ans: D

 **445. Dirty bit in paging is used to:**

- A) Check page error
- B) Track modified pages
- C) Lock a page

D) Identify invalid frames

Ans: B

 **446. Which algorithm replaces pages based on how long ago they were used?**

- A) FIFO
- B) LRU
- C) Optimal
- D) Clock

Ans: B

 **447. What is the main advantage of virtual memory?**

- A) Faster execution
- B) Allows larger programs to run
- C) Reduces fragmentation
- D) Improves CPU scheduling

Ans: B

 **448. What is the page table base register (PTBR) used for?**

- A) To point to memory blocks
- B) To indicate base address of page table
- C) To store TLB
- D) To track disk I/O

Ans: B

 **449. What is the function of the valid/invalid bit in paging?**

- A) Identify page errors
- B) Track frames
- C) Check if the page is in memory
- D) Load from disk

Ans: C

 **450. The number of page faults depends on:**

- A) CPU speed
- B) Number of processes
- C) Page replacement algorithm
- D) Disk size

Ans: C

SECTION J: Security, I/O, and Miscellaneous Concepts (451–500)

451. Which of the following is not a goal of operating system security?

- A. Confidentiality
- B. Integrity
- C. Availability
- D. Debugging

Answer: D

452. What is the main function of an I/O controller?

- A. Data processing
- B. Memory management
- C. Synchronize I/O operations
- D. Execute instructions

Answer: C

453. Which of the following is used for file protection?

- A. Firewalls
- B. Encryption
- C. Antivirus
- D. Authentication

Answer: B

454. Which of these is not an I/O technique?

- A. Programmed I/O
- B. Interrupt-driven I/O
- C. DMA
- D. Virtual I/O

Answer: D

455. What is the full form of DMA?

- A. Direct Memory Access
- B. Digital Memory Access
- C. Direct Memory Allocation
- D. Digital Machine Access

Answer: A

456. Spooling is used to:

- A. Increase CPU speed

- B. Reduce memory size
- C. Queue print jobs
- D. Swap pages

Answer: C

457. Which of the following is a valid measure for OS protection?

- A. User authentication
- B. Power backup
- C. Disk formatting
- D. Cache clearing

Answer: A

458. A trap in an OS is:

- A. Hardware failure
- B. Software-generated interrupt
- C. Timer
- D. I/O error

Answer: B

459. Which method is used to control access to resources?

- A. Mutex
- B. Paging
- C. Access matrix
- D. Segmentation

Answer: C

460. What is the use of a firewall?

- A. Schedule processes
- B. Filter network traffic
- C. Backup data
- D. Manage memory

Answer: B

461. Which among the following is not a system threat?

- A. Trapdoor
- B. Trojan horse
- C. Worm
- D. Deadlock

Answer: D

462. A virus is a:

- A. Hardware error
- B. Network protocol
- C. Malicious software
- D. Memory segment

Answer: C

463. In I/O operation, polling means:

- A. CPU waits for I/O device
- B. CPU ignores I/O request

- C. CPU schedules process
- D. Device waits for CPU

Answer: A

464. Which of these is a mechanism to avoid I/O conflicts?

- A. DMA
- B. Buffering
- C. Paging
- D. Thrashing

Answer: B

465. Which of these is not a valid I/O software layer?

- A. User-level I/O
- B. Device driver
- C. Logical I/O
- D. Physical I/O

Answer: A

466. Security breach where an attacker gets higher privilege:

- A. Privilege escalation
- B. Buffer overflow
- C. DoS
- D. Sniffing

Answer: A

467. Which of the following does not affect system security?

- A. Secure boot
- B. Password policy
- C. Process scheduling
- D. Firewalls

Answer: C

468. OS detects malware using:

- A. Timer
- B. Cache
- C. Scanner
- D. File descriptor

Answer: C

469. Which is not a hardware security measure?

- A. TPM chip
- B. BIOS password
- C. Antivirus
- D. Security lock

Answer: C

470. Which one is a type of I/O device?

- A. CPU
- B. ROM
- C. Printer

D. Cache

Answer: C

471. Daemon processes are:

- A. User processes
- B. Background services
- C. Foreground apps
- D. Virus scanners

Answer: B

472. RAID is used for:

- A. Network management
- B. Disk fault tolerance
- C. CPU efficiency
- D. RAM extension

Answer: B

473. Kernel panic is caused by:

- A. User input error
- B. Critical system failure
- C. Power failure
- D. Malware only

Answer: B

474. Time slicing helps in:

- A. Multi-threading
- B. Multitasking
- C. Multi-processing
- D. Paging

Answer: B

475. Which does not fall under OS responsibility?

- A. Process scheduling
- B. Memory management
- C. Compiler design
- D. File handling

Answer: C

476. Keylogging is a:

- A. Virus
- B. Spyware
- C. Worm
- D. Deadlock

Answer: B

477. Log files in OS are used for:

- A. Memory mapping
- B. Recording events
- C. Process scheduling

D. Disk partitioning

Answer: B

478. I/O buffering improves:

A. Security

B. File size

C. Performance

D. Authentication

Answer: C

479. Which scheduling is best for real-time systems?

A. FCFS

B. Round Robin

C. Priority

D. Rate Monotonic

Answer: D

480. Virtual machines rely on:

A. BIOS

B. Device drivers

C. Hypervisor

D. Antivirus

Answer: C

481. Which of the following is an OS utility?

A. Calculator

B. Disk defragmenter

C. Browser

D. Media player

Answer: B

482. Zombie process means:

A. Process in memory

B. Terminated but not removed

C. Running process

D. Deadlock process

Answer: B

483. OS handles deadlock using:

A. Prevention

B. Avoidance

C. Detection and recovery

D. All of the above

Answer: D

484. Secure OS allows:

A. Unauthorized access

B. Controlled data flow

C. Random execution

D. None of these

Answer: B

485. Which of these is a part of system boot process?

- A. BIOS execution
- B. Scheduler initialization
- C. Kernel loading
- D. All of these

Answer: D

486. Which file system is used in Windows?

- A. EXT3
- B. FAT32
- C. ReFS
- D. Both B and C

Answer: D

487. Real-time OS is used in:

- A. Accounting software
- B. ATM
- C. Video editing
- D. Gaming

Answer: B

488. Thrashing is related to:

- A. CPU load
- B. Disk I/O
- C. Page fault rate
- D. Boot time

Answer: C

489. Which of these is not a malicious attack?

- A. DoS
- B. Sniffing
- C. Spam
- D. Scheduling

Answer: D

490. Scheduling that selects job with least execution time:

- A. FCFS
- B. SJF
- C. RR
- D. Priority

Answer: B

491. Which utility checks disk integrity?

- A. Task manager
- B. CHKDSK
- C. FORMAT

D. REGEDIT

Answer: B

492. BIOS stands for:

- A. Binary Integrated OS
- B. Basic Input Output System
- C. Boot Input Output Setup
- D. Binary IO Setup

Answer: B

493. Disk fragmentation slows down:

- A. CPU
- B. File access
- C. Printing
- D. Booting

Answer: B

494. OS that handles multiple tasks simultaneously:

- A. Single-user
- B. Multitasking
- C. Network OS
- D. Embedded OS

Answer: B

495. A fork() system call creates:

- A. File
- B. New user
- C. New process
- D. Thread

Answer: C

496. The shell is:

- A. Hardware
- B. Interface to kernel
- C. Memory unit
- D. I/O device

Answer: B

497. UNIX OS is:

- A. Closed-source
- B. Open-source
- C. Proprietary
- D. Real-time

Answer: B

498. Which OS concept involves user-space and kernel-space separation?

- A. Paging
- B. Segmentation
- C. Protection

D. Privilege levels

Answer: D

499. Which OS is best for high-end servers?

A. Windows 10

B. Android

C. Linux

D. MS-DOS

Answer: C

500. The final job of the OS is to:

A. Schedule processes

B. Display output

C. Manage files

D. Provide user interface

Answer: D

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